Akoizo Original Practice - 2027 Rotation Form 1 - Answer key

Division B | Water Quality | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Explain why this transect is an estuary rather than a freshwater-only habitat.

Synthetic estuary transect: river end salinity 0, marine end salinity 22 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.7 to 4.7 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.8 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

answer: It connects river freshwater with seawater and has a salinity gradient produced by mixing. (2 points).

2. (2 points) Explain how a nutrient pulse can lead to lower overnight oxygen.

Synthetic estuary transect: river end salinity 0, marine end salinity 22 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.7 to 4.7 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.8 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

answer: Nutrients can increase production; respiration and decomposition consume oxygen. Nighttime photosynthesis is absent, and mixing also affects the budget. (2 points).

3. (2 points) Identify one watershed action that could reduce nutrient loading and explain its pathway.

Synthetic estuary transect: river end salinity 0, marine end salinity 22 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.7 to 4.7 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.8 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

answer: For example, reduce excess fertilizer and retain vegetated buffers to limit nutrient transport in runoff; identify a source and transport pathway. (2 points).

4. (2 points) Explain how traditional ecological knowledge could improve this monitoring plan without replacing measured evidence.

Synthetic estuary transect: river end salinity 0, marine end salinity 22 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.7 to 4.7 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.8 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

answer: Collaborate with knowledge holders to identify historical baselines, seasonal changes and appropriate sampling locations; evaluate observations alongside recorded measurements and respect consent/context. (2 points).

5. (2 points) Why should a reef food-web analysis distinguish herbivores from predators?

answer: They affect different trophic pathways; grazing regulates algae while predation affects prey populations, with indirect effects across the web. (2 points).

6. (2 points) Distinguish a harmful algal bloom from an ordinary observation of green water.

answer: A bloom can cause toxins or oxygen stress, but color alone does not establish harmful species, toxins or ecological damage; additional evidence is needed. (2 points).

7. (2 points) A reef fish has fused beak-like teeth and scrapes algae from hard surfaces. Identify its common-name group.

answer: Parrotfish (2 points).

8. (2 points) A starfish has many arms and venomous spines and consumes live coral tissue. Identify it.

answer: Crown of thorns starfish (2 points).

9. (2 points) A reef-building animal secretes a hard calcium carbonate skeleton. Identify its listed group.

answer: Hard/stony coral (2 points).

10. (2 points) A dark urchin has conspicuously long slender spines and grazes algae. Identify its listed group.

answer: Long-spined black sea urchin (2 points).

11. (2 points) A shrimp has red-and-white bands and conspicuous elongated appendages. Identify the listed common-name group.

answer: Banded coral shrimp (2 points).

12. (2 points) An elongate, soft-bodied echinoderm processes sediment on the seafloor. Identify its listed group.

answer: Sea cucumber (2 points).

13. (2 points) Calculate the overnight dissolved-oxygen decline.

Synthetic estuary transect: river end salinity 0, marine end salinity 22 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.7 to 4.7 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.8 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

answer: 3 mg/L (2 points).

Numerical target 3; absolute tolerance 0.006. Accepted units: , mg/L.

14. (2 points) Calculate the BOD-bottle oxygen demand under the stated assumptions.

Synthetic estuary transect: river end salinity 0, marine end salinity 22 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.7 to 4.7 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.8 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

answer: 3.2 mg/L (2 points).

Numerical target 3.2; absolute tolerance 0.0064. Accepted units: , mg/L.

15. (2 points) A mixture contains equal volumes of freshwater and seawater with salinity 22. Assuming additive volumes and conservative mixing, estimate salinity.

Synthetic estuary transect: river end salinity 0, marine end salinity 22 practical-salinity units. After a nutrient pulse, dissolved oxygen falls from 7.7 to 4.7 mg/L overnight. A separate closed BOD bottle starts at 8.0 mg/L and ends at 4.8 mg/L; ignore dilution and nitrification. These are teaching observations, not measurements from a real reef.

answer: 11 (2 points).

Numerical target 11; absolute tolerance 0.022. Accepted units: , .

16. (2 points) Aragonite saturation decreases while other factors are held fixed. Explain a relevant concern for stony corals.

answer: Lower aragonite saturation can make calcium-carbonate skeleton formation less favorable and increase dissolution risk; responses depend on organism and conditions. (2 points).

17. (2 points) Explain why high turbidity and high dissolved-solids concentration are not interchangeable observations.

answer: Turbidity reflects light scattering by suspended material. Dissolved substances can change chemistry without producing the same scattering signal. (2 points).

18. (2 points) A field pH value changes between daytime and predawn. Name a biological mechanism consistent with this and one check on the inference.

answer: Photosynthesis/respiration change dissolved CO2 and can shift pH. Check timing, calibration, temperature, depth, mixing and repeated observations before assigning a cause. (2 points).

19. (2 points) A salinometer calibration gives reading 10 at 2% mass/volume salt and reading 30 at 6%. A sample reads 16. Use linear interpolation to estimate concentration.

answer: 3.2 % (2 points).

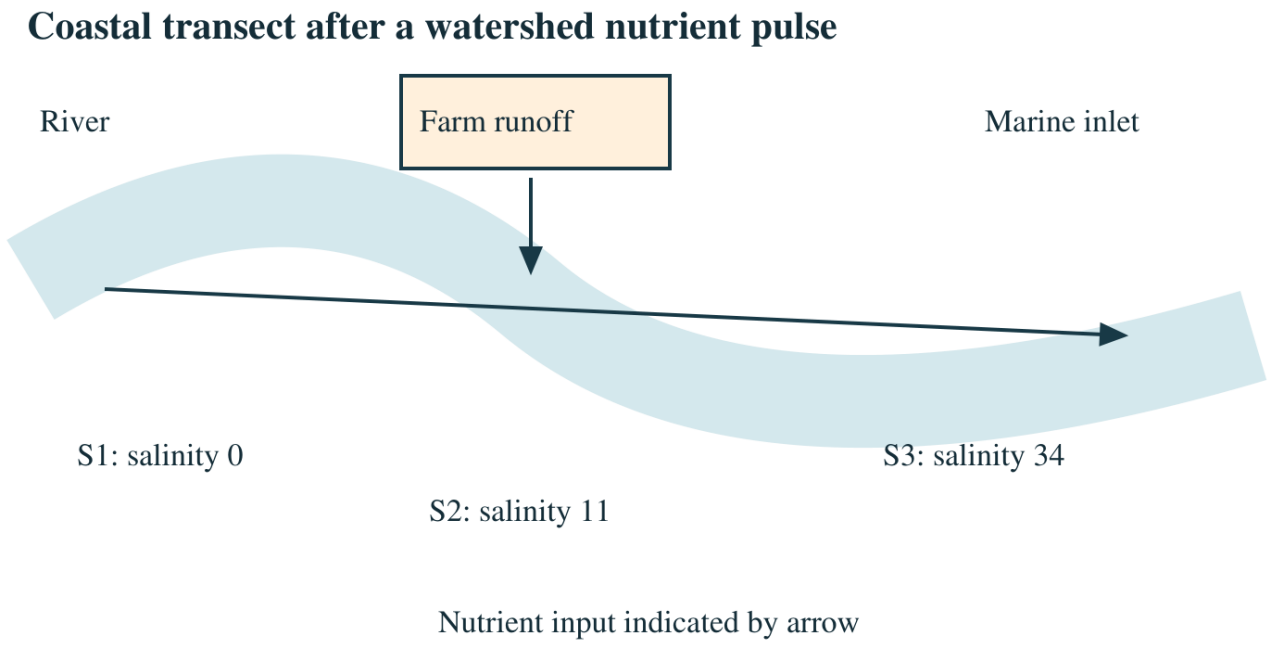
Numerical target 3.2; absolute tolerance 0.0064. Accepted units: , %.

20. (2 points) How many grams of salt are required for 500 mL of final solution at 3% mass/volume?

answer: 15 g (2 points).

Numerical target 15; absolute tolerance 0.03. Accepted units: , g.

21. (4 points) Use Figure F1. Which labeled station most clearly represents brackish mixing rather than either freshwater or full seawater? Explain.



F1. Estuary sampling map (original synthetic teaching figure)

answer: S2; its intermediate salinity is consistent with freshwater-seawater mixing. (4 points).

22. (4 points) Use Figure F1. Trace a plausible route from the mapped nutrient source to reef oxygen stress.

Refer to F1. Estuary sampling map (original synthetic teaching figure) shown with the first question in this figure group.

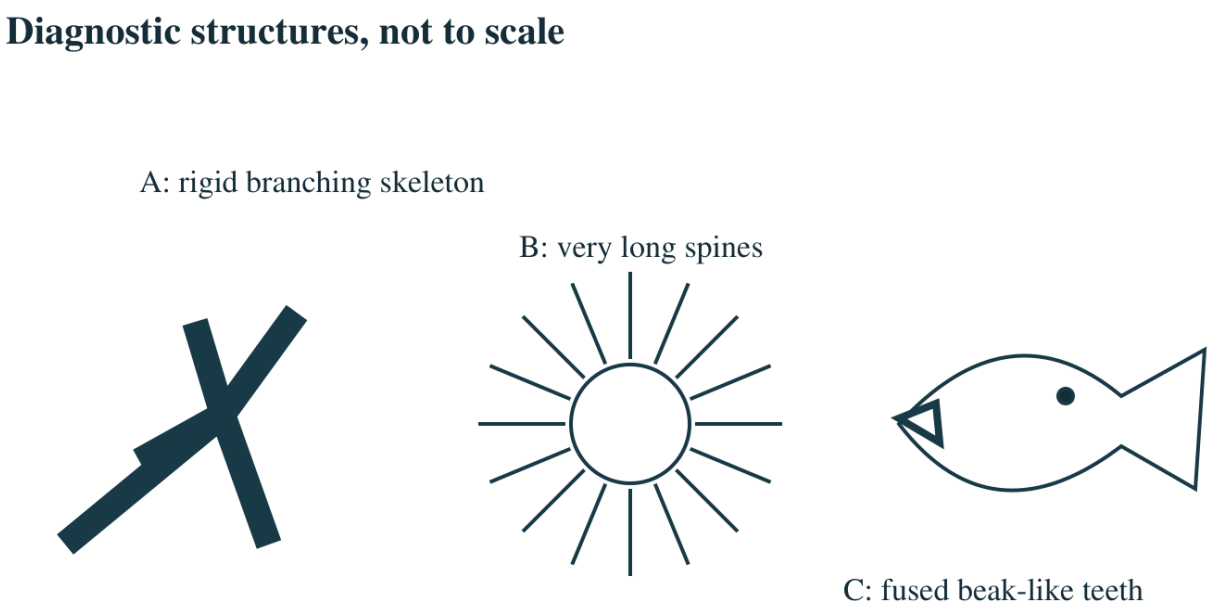
answer: Runoff can transport nutrients to the estuary and coast; enhanced production and subsequent respiration/decomposition can consume oxygen. The map alone does not prove the full causal chain. (4 points).

23. (4 points) Use Figure F1. Place a vegetated buffer at an appropriate location on the map and explain its purpose.

Refer to F1. Estuary sampling map (original synthetic teaching figure) shown with the first question in this figure group.

answer: Between the farm/runoff source and the channel, to intercept runoff and reduce nutrient transport before it reaches the estuary. (4 points).

24. (4 points) Use Figure F2. Identify the listed reef-building group represented by A.



F2. Reef organism identification sketches (original synthetic teaching figure)

answer: Hard/stony coral. (4 points).

25. (4 points) Use Figure F2. Identify the listed urchin group represented by B.

Refer to F2. Reef organism identification sketches (original synthetic teaching figure) shown with the first question in this figure group.

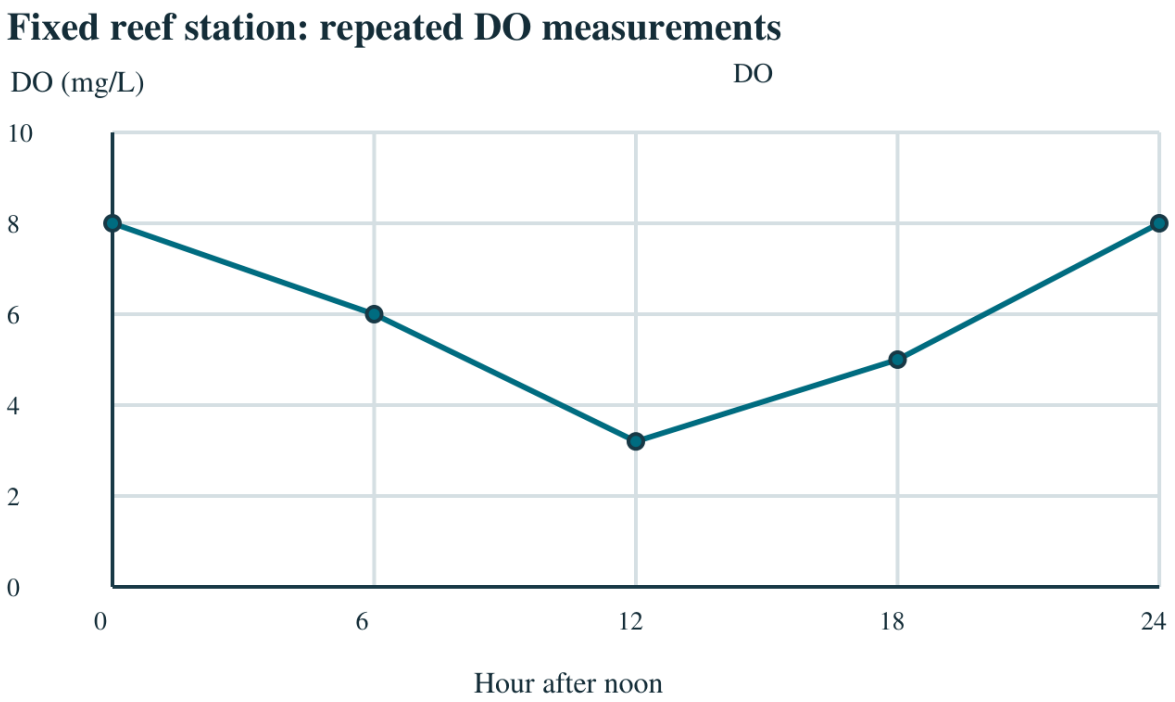
answer: Long-spined black sea urchin. (4 points).

26. (4 points) Use Figure F2. Identify C and explain how its feeding could affect competition for reef space.

Refer to F2. Reef organism identification sketches (original synthetic teaching figure) shown with the first question in this figure group.

answer: Parrotfish; algal grazing can reduce competition between macroalgae and corals, although ecosystem responses depend on other conditions. (4 points).

27. (4 points) Use Figure F3. Read the lowest observed DO concentration.



F3. Diel dissolved oxygen (original synthetic teaching figure)

answer: 3.2 mg/L (4 points).

Numerical target 3.2; absolute tolerance 0.0096. Accepted units: , mg/L.

28. (4 points) Use Figure F3. Calculate the observed maximum-to-minimum DO range.

Refer to F3. Diel dissolved oxygen (original synthetic teaching figure) shown with the first question in this figure group.

answer: 4.8 mg/L (4 points).

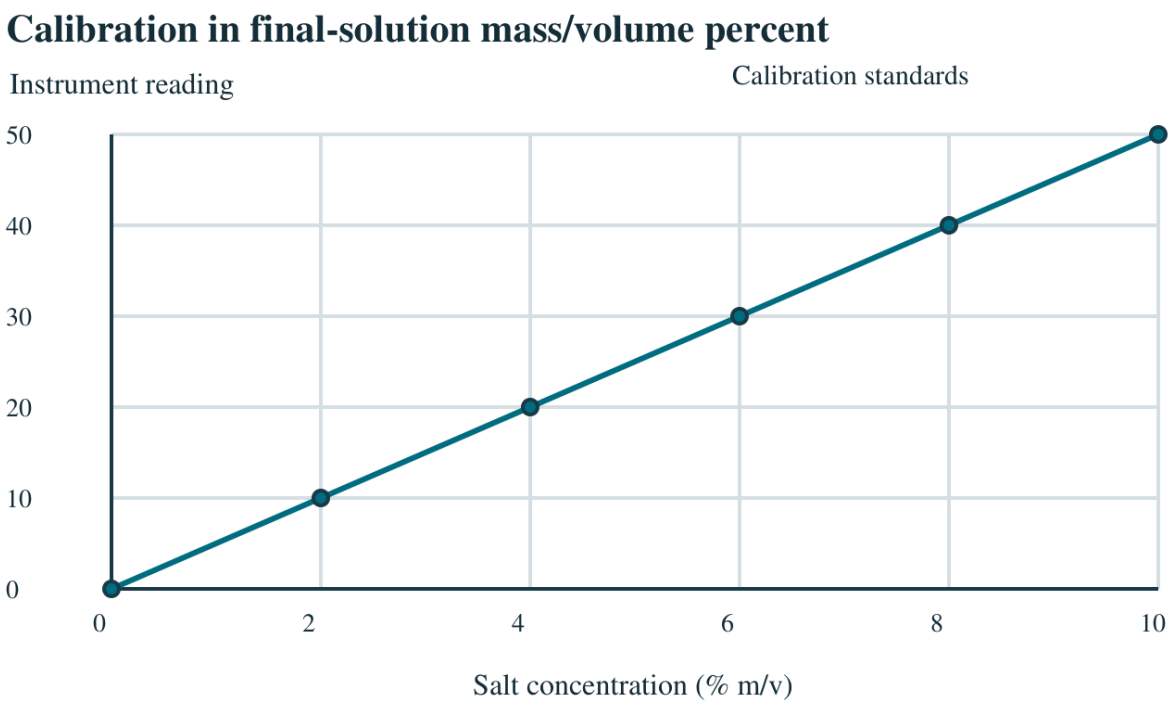
Numerical target 4.8; absolute tolerance 0.0144. Accepted units: , mg/L.

29. (4 points) Use Figure F3. Why would a noon-only survey miss information visible in this graph?

Refer to F3. Diel dissolved oxygen (original synthetic teaching figure) shown with the first question in this figure group.

answer: It would miss lower overnight oxygen and the daily variation, so the repeated time series is needed to characterize exposure. (4 points).

30. (2 points) Use Figure F4. A sample gives reading 21. Interpolate its salt concentration.



F4. Salinometer calibration (original synthetic teaching figure)

answer: 4.2 % (2 points).

Numerical target 4.2; absolute tolerance 0.0126. Accepted units: , %.

31. (1 points) Use Figure F4. How many grams of salt are needed for 200 mL final solution at the 4% standard?

Refer to F4. Salinometer calibration (original synthetic teaching figure) shown with the first question in this figure group.

answer: 8 g (1 points).

Numerical target 8; absolute tolerance 0.024. Accepted units: , g.

32. (1 points) Use Figure F4. Why is extrapolating this line to a reading of 80 less supported than interpolation?

Refer to F4. Salinometer calibration (original synthetic teaching figure) shown with the first question in this figure group.

answer: Reading80 lies outside the tested calibration range; linear behavior there has not been demonstrated. (1 points).